

Business Requirements Document (BRD)

E-commerce Sales Performance Pipeline

Document Control

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- Status: Draft v1.0
- Data note: Built on a cleaned/simulated order dataset processed in Google Colab, created for a portfolio application; not live production data.

1.Purpose

This document defines the business requirements for a customer value and retention analytics pipeline. The goal is to help the marketing and sales ops teams understand which customers are most valuable and how well the business retains customers over time, replacing manual, spreadsheet-based customer analysis.

2.Business Objectives

- Segment customers by value (Recency, Frequency, Monetary) so marketing can prioritize retention offers toward high-value segments.
- Track repeat-purchase behavior by acquisition cohort to see whether retention is improving or declining month over month.
- Give a non-technical stakeholder a single Excel/Power BI view instead of raw SQL exports.

3.Scope

In Scope

- Cleaning and structuring raw order data into a normalized schema (customers, products, orders, order_items).
- RFM segmentation of customers into value tiers (e.g. Champions, Loyal, At Risk, Lost).

- Cohort retention analysis showing repeat-purchase. rate by acquisition month.
- Power BI dashboard connected live to MySQL, plus an Excel summary workbook for stakeholders

Out of Scope

- Executing marketing campaigns or offers based on the segments (analysis only, not activation).
- Real-time/streaming data updates.
- Payment processing or fraud detection.

4.Stakeholders

- Marketing Manager — primary consumer of RFM segments; uses them to target retention campaigns.
- Sales Ops — consumer of cohort retention trends; uses them to track overall account health.
- Founder/Business Owner — secondary consumer; wants a simple summary view, not raw data.

5.Functional Requirements

ID	Requirement	Priority	Acceptance Criteria
FR-1	System shall clean raw order data and load it into a normalized MySQL schema (customers, products, orders, order_items).	High	Given the raw dataset, when the Colab cleaning pipeline runs, then all four tables exist in MySQL with no duplicate order_items and no null customer_id values.
FR-2	System shall calculate RFM scores for every customer and assign a segment label.	High	Given the orders table is populated, when the RFM script runs, then every customer has a Recency, Frequency, Monetary score and one segment label.
FR-3	System shall display a cohort retention view showing repeat-purchase rate by acquisition month.	High	Given at least two months of order data, when the cohort view renders, then repeat-purchase rate is shown for each acquisition month cohort across subsequent months.

ID	Requirement	Priority	Acceptance Criteria
FR-4	System shall connect the Power BI dashboard live to the MySQL database.	Medium	Given new orders are added to MySQL, when the Power BI dashboard is refreshed, then updated totals appear without manual re-import.
FR-5	System shall produce an Excel summary workbook translating RFM segments and cohort trends into a stakeholder-readable view.	Medium	Given the analysis is complete, when the Excel workbook is opened, then segment counts and the retention trend are visible without needing to read SQL or Python.

6.Non-Functional Requirements

- Reproducibility: The Colab notebook must run end-to-end on a fresh copy of the raw dataset without manual intervention.
- Usability: Excel/Power BI outputs must be interpretable by a non-technical stakeholder.
- Maintainability: MySQL schema must be documented (column names, types, relationships) for future extension.

7.Assumptions & Constraints

- Dataset is a cleaned/simulated e-commerce order export, not a live payment or inventory system feed.
- RFM segment thresholds are defined once for this analysis and not dynamically re-tuned.
- Google Colab is used for one-time/batch processing, not a scheduled production pipeline.

8.Acceptance Sign-off Criteria

This phase is complete when: the MySQL schema is populated (FR-1), every customer has an RFM segment (FR-2), the cohort retention view renders correctly (FR-3), the Power BI dashboard reflects live MySQL data (FR-4), and the Excel summary (FR-5) is reviewed and understood by a non-technical stakeholder persona without walkthrough.